

ShiftGrid: Product Specification & Development Blueprint

Building a Self-Hostable Workforce Management Solution

1. Executive Summary & Vision

ShiftGrid is an open-source, self-hostable workforce management platform designed to replace fragmented scheduling methods used by small-to-medium enterprises (SMEs). Our vision is to democratize high-end scheduling software, providing cafes, local healthcare clinics, and small retail shops with a professional-grade tool that respects data sovereignty and budget constraints.

Unlike expensive SaaS alternatives that charge per-user fees, ShiftGrid empowers business owners to host their own infrastructure while benefiting from a community-driven feature set. We aim to transform "the spreadsheet struggle" into a streamlined, automated, and mobile-friendly experience.

2. Target Audience & Market Problem

Small business operators frequently manage human resources through a chaotic mix of messaging apps, physical whiteboards, and static spreadsheets. This leads to several critical pain points:

- **Communication Chaos:** Shift swap requests buried in group chats often lead to no-shows or double-booking.
- **Compliance Risks:** Difficulty in tracking hours leads to accidental labor law violations regarding mandatory break times and overtime thresholds.
- **Administrative Overhead:** Managers spend hours manually calculating availability and verifying swap eligibility.
- **Inaccessibility:** Staff cannot check their schedules in real-time from mobile devices without calling the workplace.

3. Core Feature Set

ShiftGrid focuses on high-utility features that solve the most immediate friction points for staff and managers.

Interactive Schedule Management

- **Weekly/Monthly Grid:** A responsive drag-and-drop interface for managers to assign shifts and visualize coverage.
- **Role-Based Views:** Staff see their personal schedule; managers see the entire organization.

Peer-to-Peer Shift Swapping

- **Swap Requests:** Employees can offer their shifts to a marketplace or specific colleagues.
- **Manager Approval Workflow:** Automatic conflict checking ensures the swap doesn't violate overtime rules before the manager receives a final approval prompt.

Real-time Connectivity

- **Web Push Notifications:** Immediate alerts for new shifts, schedule changes, or swap approvals.
- **In-App Inbox:** A centralized location for all schedule-related correspondence.

Time & Attendance

- **Digital Clock-In:** Location-verified (optional) check-in/out via the mobile web interface.
- **Automatic Overtime Flagging:** System-level alerts when a staff member approaches local labor law limits.

4. Developer Lead Responsibilities & Architecture

The Technical Lead is responsible for maintaining the integrity of the open-source codebase and ensuring high performance in self-hosted environments.

Technology Stack

Layer	Technology
Frontend	Next.js (App Router), React, Tailwind CSS
Backend	Node.js (Express/Fastify) or FastAPI (Python)
Database	PostgreSQL
State Management	React Query / TanStack Query
Deployment	Docker & Docker Compose

Database Schema Requirements

- **Users Table:** Standardizing roles (Admin, Manager, Staff), hourly rates, and contact info.
- **Shifts Table:** Start/end times, role requirements, and location metadata.
- **Swap_Requests Table:** Tracking the status (Pending, Offered, Accepted, Approved, Rejected) of shift transfers.
- **Attendance Table:** Log of actual vs. scheduled hours for payroll reconciliation.

Infrastructure & CI/CD

- Standardized Docker containers for "One-Click" deployment via platforms like Railway, Fly.io, or on-premise servers.
- GitHub Actions for automated testing and linting to maintain code quality across community contributions.

5. Business Lead Responsibilities

The Business Lead focuses on market fit, legal frameworks, and community growth.

- **Customer Discovery:** Conducting interviews with at least 10 local business owners to validate feature prioritization.
- **Compliance Framework:** Drafting a "Labor Law Configurator" that allows users to set rules based on their specific jurisdiction (e.g., California OT vs. EU Working Time Directive).
- **Open-Source Management:** Curating the GitHub README, managing the public roadmap, and fostering a welcoming environment for contributors.
- **Documentation:** Creating user-facing guides for setup and day-to-day use.

6. 4-Week MVP Roadmap & Launch Strategy

The goal is to move from concept to a deployable GitHub repository in 30 days.

Week	Phase	Focus Areas
Week 1	Foundation	Setup DB schema; Auth (NextAuth.js); Basic Shift Grid UI.
Week 2	Core Logic	Implementation of Shift Creation and Basic Swap Request flow.
Week 3	Notifications	Integration of real-time alerts; Role-based access control hardening.
Week 4	Launch Prep	Dockerization; Documentation; Public GitHub Launch.

GitHub Launch Strategy

1. **Release v0.1-Alpha:** Tag a stable build with a focus on the core "Schedule & Swap" functionality.

2. **Product Hunt Launch:** Schedule a launch for  **Date** to drive initial open-source awareness.
3. **Community Call-to-Action:** Encourage developers to contribute "Localization Modules" for regional labor laws.

Project Lead:  **Person**

Primary Repository:  **File**